

Review

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Proposed revision of 'crystal cells' as phagocytised metabolic by-products in holothuroid echinoderms

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Abstract

Crystal cells in echinoderms have been described in the literature as a type of coelomocyte (immune cell) containing a crystalline structure. Their putative function has been widely ascribed to osmoregulation but this assertion was never robustly tested. In the present paper, a review of crystal cells and crystalline structures provides evidence supporting a different identity and function. First, the same microcrystals can be found either free or encapsulated by a cell membrane (with or without a structure resembling a nucleus). Specifically, they are typically non-encapsulated when found inside tissues of internal organs and encapsulated when free floating in the hydrovascular or perivisceral fluids. Although usually individually packaged, microcrystals were also observed encapsulated in groups of up to four, with or without other particles. Their morphological features, coupled with their chemical and optical properties, match that of microcrystals of uric acid, described in other phyla, including chordates. Two pathways of excretion of these crystallised by-products were evidenced: rejected with sea water out of the respiratory tree and expelled via transrectal coelomoducts among coelomocyte aggregates. Overall, the present synthesis strongly supports that 'crystal cells' historically described as a distinct type of coelomocyte in holothuroids are phagocytes that have engulfed uric acid microcrystals generated as waste by metabolic activities.

Introduction

Various types of coelomocytes in echinoderms have been described and are regarded as multi-functional cells, mostly serving as part of an arsenal of immune cells. The identification of these coelomocytes has conventionally been based on morphology as viewed through microscopy (Chia and Xing, 1996; Queiroz *et al.*, 2021; Queiroz and Custódio, 2024). Among the different classes in the phylum, there is a relatively high diversity of coelomocytes in holothuroids with at least nine recognised types: phagocytes (also known as leukocytes), spherule cells (also known as spherulocytes, amoebocytes, morula cells, or granulocytes), fusiform cells, hemocytes, giant cells, progenitor cells, minute corpuscles (fragments of cells), lymphocytes (small spherical cells), and crystal cells (crystalline cells or crystal corpuscles) (Endean, 1966; Smith, 1981; Chia and Xing, 1996; Smith *et al.*, 2018; Queiroz *et al.*, 2021; Queiroz and Custódio, 2024). Previously, vibratile cells were considered a type of coelomocytes in holothuroids until a recent analysis demonstrated that they are in fact male gametes at different stages of development (Caulier *et al.*, 2024).

Among the coelomocytes, crystal cells have mostly been described in holothuroids (Smith, 1981; Smith *et al.*, 2018) and have also been mentioned in echinoids and asteroids (Johnson and Beeson, 1966; Smith, 1981; Chia and Xing, 1996; Queiroz *et al.*, 2022a). They were observed in 40% of ~80 studied species of holothuroids, including in up to 49% of the Dendrochirotida investigated (Queiroz and Custódio, 2024). In this context, crystal cells are thought to be highly differentiated coelomocytes (Zavalnaya *et al.*, 2020). In most echinoderm taxa, crystal cells typically comprise between 0.5 and 5% of total coelomocyte cell counts (Endean, 1958; Smith, 1981; Chia and Xing, 1996; Smith *et al.*, 2018; Queiroz *et al.*, 2021, 2022a, b; Jobson *et al.*, 2022; Xue *et al.*, 2024). In some instances, the proportion of crystal cells can be as high as ~10% of all coelomocytes (Caulier *et al.*, 2020). Although not always present in a given individual, their densities in the coelomic fluid of holothuroids average around 600 cells mL⁻¹ (Eliseikina and Magarlamov, 2002) but can reach upwards of 13,500 cells mL⁻¹ (Jobson *et al.*, 2021) and 143,000 cell mL⁻¹ (Li *et al.*, 2018). Crystal cells are usually ascribed a crescent-shaped, heterochromatic nucleus and an intracellular vacuole containing the crystalline structure (Xing *et al.*, 2008; Smith *et al.*, 2018; Caulier *et al.*, 2020; Jobson *et al.*, 2022; Queiroz and Custódio, 2024; Xue *et al.*, 2024). However, the nucleus is often indistinguishable, which suggests that it is either absent or, if present, challenging to identify (Hetzl, 1963; Chia and Xing, 1996; Chia and Xing, 1996; Eliseikina and Magarlamov, 2002; Caulier *et al.*, 2020).

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While the function of crystal cells in echinoderms was generally considered unknown (Ratcliffe and Rowley, 1979; Smith, 1981; Karp and Coffaro, 1982; Smiley, 1994; Smith et al., 2018), some authors have proposed they relate to mesenchymal ossicle-forming cells (Hetzel, 1963) or play a role in osmoregulation (Eliseikina and Magarlamov, 2002; Xing et al., 2008; Ramírez-Gómez and García-Arrarás, 2010). Hetzel (1963) noted that a slight change in osmotic pressures can cause the disintegration of the intracellular crystalline structure in crystal cells (Hetzel, 1963; Endean, 1966; Chia and Xing, 1996). Upon dissolution of the crystalline structure, the remaining cell collapses (Hetzel, 1963). Eliseikina and Magarlamov (2002) observed that an increase or normalisation in osmotic pressure could result in the transformation of crystal cells into structurally similar cells (i.e., cells with a centrally located spherical vacuole), suggesting that the crystallisation process inside the cell was reversible (also see Xing et al., 2008). However, any role with osmoregulation has been recently challenged when the density of crystal cells did not change under experimental exposure to low salinities (Jobson et al., 2021). The solubility of the crystal-like structure under a slight change in osmotic pressure and its refractive properties have led several authors to conclude that the structure is crystalline in nature (Endean, 1966; Karp and Coffaro, 1982; Chia and Xing, 1996) and, more recently, that its chemical identity is that of uric acid (KCK et al., 2025). Briefly, crystals of uric acid in holothuroids were identified based on their morphology (cuboidal-barrel-, rhomb-, and rosette-shaped crystals) and size ranges, solubility in alkaline solution or upon heating but not in alcohol, and diagnostic Raman peaks at 839, 995, 1036, 1158, and 1649 cm^{-1} (KCK et al., 2025).

In line with emerging questions surrounding some coelomocyte types (Karp and Coffaro, 1982; Taguchi et al., 2016; Caulier et al., 2024; Wambreuse et al., 2025), the present review revisits the identity of crystal cells as a distinct type of coelomocyte. First, it summarises available information on the so-called crystal cells among coelomocytes of holothuroids and other echinoderms (e.g., asteroids, echinoids) and, second, reassesses this notion on the basis of a recent discovery and analysis of microscopic crystalline structures (i.e., microcrystals) in holothuroids (KCK et al., 2025).

Review of crystal-cell coelomocytes in echinoderms

Morphology of crystal cells

Rhomb-shaped crystal cells

Crystal cells with a single rhomb-shaped crystalline structure have been documented in asteroids and holothuroids (Table 1). The shape of this crystal has been described as rhomboid(al), rhomb(ic), lozenge-shaped, or diamond-shaped (Table 1). In particular, cells have been described with two distinct rhomboidal crystals (Endean, 1958), three distinct crystals (Endean, 1966; Smith, 1981; Smiley, 1994), and an arrangement of three individual crystals overlaid at offset angles that can give an appearance of a six-pointed star-shaped morphology (Hetzel, 1963).

The sizes of these crystal cells in holothuroids vary, with studies reporting lengths of 2–4 μm (Smiley, 1994) and larger sizes of 7–8 μm in length and 4–5 μm in width (Chia and Xing, 1996; Xing, 1997; Xing et al., 1998) as well as 5–6 μm in length (maximum of 24 μm), 4–5 μm in width (maximum of 16 μm), and 1–2 μm in thickness (maximum of 11 μm) (Hetzel, 1963, 1965) (Table 2). Species-specific measurements include lengths of 12–30 μm in *Paraleptopentacta elongata* and *Panningia hyndmani*, 15 μm in *Parastichopus tremula*, and 22–28 μm in *Ocnus lactea* (Théel, 1921;

Table 2). In *Molpadia roretzii*, they measured 6–8 μm in length, 3–5 μm in width, and 2–8 μm in thickness (Ohuye, 1936; Table 2). In *Holothuria glaberrima*, they had an average length of 5 μm (Ramírez-Gómez et al., 2010; Table 2). In *Holothuria leucospilota*, they measured 9–11 μm long, 7–9 μm wide, and 4–5 μm thick (Endean, 1958; Table 2). In *Cucumaria frondosa*, they ranged between 15 and 22 μm in length (Caulier et al., 2020; also see illustrations by Jobson et al., 2021, 2022; Table 2). Length measurements of crystal cells in holothuroids, likely to closely represent the length of the crystal, may or may not include the cytoplasm and/or nucleus described in some publications; reports have ranged from 2 to 24 or 4 to 24 μm (Smith et al., 2018; Queiroz and Custódio, 2024; Tables 1 and 2). The size of rhomb-shaped crystal cells in asteroids has not been reported.

Cuboidal crystal cells

Crystal cells with cuboidal crystalline structures have been reported in crinoids, echinoids, ophiuroids, and holothuroids (Cuénot, 1891; Xing et al., 2008; Chen et al., 2015; Queiroz et al., 2022a). Among holothuroids, these crystals are less frequently observed compared to rhomb-shaped ones (described above). In the literature, this morphology has been described variously as cuboid(al), cubic, or cube-shaped (Table 1). In one instance, the shape was referred to as cuboid prismatic (Cuénot, 1891), but it is unclear if other mentions of prisms/prismatic forms also refer to the cuboidal shape. Notably, the surfaces of some cuboidal crystals were slightly curved (Cuénot, 1891). In the echinoid *Paracentrotus gaimardi*, live cells with this type of crystalline structure, including the surrounding cytoplasm, measured approximately 5 μm in diameter (Queiroz et al., 2022a; Table 2). The sizes of those observed in echinoids by Cuénot (1891) were only described as variable. Based on a published image, the cuboidal crystal in the holothuroid *Apostichopus japonicus* was ~6 μm in length (Xing et al., 2008; Chen et al., 2015; Table 2).

Other morphologies of crystal cells

Star-shaped/stellate, rosette-shaped, cross-shaped, hexagonal (also referred to as polyhedrons), rod-shaped/elongated/styloid/needle-like, rectangular, squarish, and spherical/round crystalline structures have been reported in asteroids and holothuroids (Table 1). Regular prisms and prismatic shapes in crinoids, echinoids, and holothuroids are too generic to distinguish them from or classify them with the other named morphologies (Cuénot, 1891; Xue et al., 2024; Wambreuse et al., 2025). Théel (1921) remarked that rectangular crystals may be the side view of rhomb-shaped crystals and that cubic crystals may be the side view of star-shaped crystals. There are also reservations of the actual morphology of quadrilateral crystals in *Apostichopus japonicus*, as they may be either rhombic or rectangular crystals (Liang et al., 2019). There are no instances of multiple crystals within a cell reported for any of these shapes (Table 1).

Star-shaped/stellate crystalline structures are generally not as common compared to rhomb-shaped ones (Endean, 1966). They have been reported in the holothuroid genera *Psolus* and *Psolidium* with diameters up to 70 μm (Hetzel, 1963, 1965; Endean, 1966; Table 2). In *Psolus chitonoides*, this crystalline structure, based on an illustration, measures 12 μm in diameter (Hetzel, 1963, 1965; Table 2). Additionally, this crystal shape is known in other holothuroid taxa – namely, *Panningia hyndmani*, *Paraleptopentacta elongata*, and *Parastichopus tremula* – with diameters ranging from 11 to 19 μm (Théel, 1921; Table 2). The cross-shaped crystal isolated

Table 1. Summary of the diversity of morphological descriptions of crystal cells identified as a type of coelomocyte in echinoderms, organised by class and then by year of publication

Taxon	Shape	Size	Reference
<u>Crinoidea</u>			
Unspecified	Cuboid, regular prism	No data	Cuénot (1891)
<u>Asteroidea</u>			
<i>Patiria miniata</i>	Fusiform, lozenge-shaped, almost needle-like	No data	Johnson and Beeson (1966)
Unspecified	No description	No data	Ratcliffe and Rowley (1979)
<u>Ophiuroidea</u>			
Unspecified	Cuboid prismatic	No data	Cuénot (1891)
<u>Echinoidea</u>			
For example: <i>Strongylocentrotus lividus</i>	Cubes	No data	Cuénot (1891)
<i>Paracentrotus gaimardi</i>	Cubic	Diameter: ~5 µm (including cytoplasm)	Queiroz <i>et al.</i> (2022a)
Unspecified	No description	No data	Xue <i>et al.</i> (2024)
<u>Holothuroidea</u>			
<i>Holothuria coluber</i> , <i>Phyllophorus cebuensis</i> (previously as <i>Thyonidium cebuense</i>), <i>Stolus canescens</i> (previously as <i>Cucumaria canescens</i>)	Diamond, cross	No data	Semper (1868)
<i>Mesothuria intestinalis</i> , <i>Ocnus lacteus</i> (previously as <i>Cucumaria lactea</i>), <i>Panningia hyndmani</i> (previously as <i>Cucumaria hyndmani</i>), <i>Paraleptopentacta elongata</i> (previously as <i>Cucumaria elongata</i>), <i>Parastichopus tremula</i> (previously as <i>Stichopus tremula</i>), <i>Psolus phantapus</i> , <i>Thyone fusus</i>	Rhombic (also biconvex rhomboid) or its derivatives (star-shaped); rectangular rod or quadrangular prism	Length: 22.0–27.8 µm (rhombic, <i>O. lacteus</i>); length: 11.9–30.4 µm (rhombic, <i>P. elongata</i> and <i>P. hyndmani</i>); length: 15.2 µm (rhombic, <i>P. tremula</i>), diameter: 11.4–19.0 µm (star-shaped, <i>P. elongata</i> and <i>P. hyndmani</i>); diameter: 11.4 µm (star-shaped, <i>P. tremula</i>); length: 7.6–15.2 µm (quadrangular prism, <i>M. intestinalis</i>)	Théel (1921)
<i>Psolus phantapus</i>	No description	No data	Kindred (1924)
<i>Caudina chilensis</i> , <i>Molpadia roretzii</i>	Rhomboidal	Length: 6–8 µm (<i>M. roretzii</i>); width: 3–5 µm (<i>M. roretzii</i>); thickness: 2–8 µm (<i>M. roretzii</i>)	Ohuye (1936)
<i>Holothuria leucospilota</i>	Rhomboidal	Length: 9–11 µm; width: 7–9 µm; thickness: 4–5 µm	Endean (1958)
Unspecified	Rhomboidal	No data	Booolootian and Giese (1958), Booolootian (1962)
Unspecified	Spindle-shaped, long fusiform	No data	Andrew (1962)
<i>Apostichopus californicus</i> (previously as <i>Parastichopus californicus</i>), <i>Cucumaria lubrica</i> , <i>Cucumaria miniata</i> , <i>Cucumaria piperata</i> , <i>Eupentacta pseudoquinquesemita</i> , <i>Eupentacta quinquesemita</i> , <i>Molpadia intermedia</i> , <i>Pentamera populifera</i> , <i>Psolidium bullatum</i> , <i>Psolus chitonoides</i>	Rhomboidal, imperfect star-shaped	Length: mean of 5–6 µm, up to 24 µm (rhomboidal); width: mean of 4–5 µm, up to 16 µm (rhomboidal); thickness: mean of 1–2 µm, up to 11 µm (rhomboidal); diameter: 11.9 µm (imperfect star-shaped determined from image, <i>P. chitonoides</i>); diameter: up to 70 µm (imperfect star-shaped, <i>Psolus</i> spp. and <i>Psolidium</i> spp.)	Hetzel (1963, 1965)
Unspecified	Stellate	No data	Endean (1966)
Unspecified	No description	No data	Ratcliffe and Rowley (1979)
<i>Apostichopus californicus</i> (previously as <i>Parastichopus californicus</i>)	Hexagonal, rod-shaped	No data	Dybas and Fankboner (1986)

(Continued)

Table 1. (Continued.)

Taxon	Shape	Size	Reference
Unspecified	Rhomboid	Length: 2–4 μm	Smiley (1994)
<i>Apostichopus japonicus</i> , <i>Cucumaria japonicus</i>	Rectangular	Length: 6–7 μm ; width: 4 μm	Eliseikina and Magarlamov (2002)
<i>Apostichopus japonicus</i>	Hexagonal	Diameter: ~ 5 μm	Xing and Yang (2008)
<i>Apostichopus japonicus</i>	Hexagonal, styloid, cuboid	Diameter: 8.0 μm (hexagonal, determined from image); length: 15.6 μm (styloid, determined from image); length: 5.9 μm (cuboid, determined from image)	Xing <i>et al.</i> (2008), Chen <i>et al.</i> (2015)
<i>Apostichopus japonicus</i>	Polyhedron	Length: 5–10 μm	Li <i>et al.</i> (2009)
Unspecified	Rhomboidal, hexagonal	No data	Ramírez-Gómez <i>et al.</i> (2010)
<i>Holothuria glaberrima</i>	Rhomboidal	Length: mean of 5.4 μm (including cytoplasm)	Ramírez-Gómez <i>et al.</i> (2010)
<i>Apostichopus japonicus</i>	No description	Length: 3.6 μm (unspecified shape, determined from image)	Li <i>et al.</i> (2013)
Unspecified	Rhomboid	Length: 2–24 μm	Smith <i>et al.</i> (2018)
<i>Apostichopus japonicus</i>	Hexagonal	Diameter: 7.4 μm (determined from image)	Li <i>et al.</i> (2018)
<i>Apostichopus japonicus</i>	Hexagons, quadrilaterals, spheres	Length: 5–10 μm (unspecified shape)	Liang <i>et al.</i> (2019)
<i>Cucumaria frondosa</i>	Diamond	Length: 15–22 μm	Caulier <i>et al.</i> (2020)
<i>Eupentacta fraudatrix</i>	No description	No data	Zavalnaya <i>et al.</i> (2020)
<i>Cucumaria frondosa</i>	No description	Length: 20.9 μm (rhombic, determined from image)	Jobson <i>et al.</i> (2021), (2022)
<i>Holothuria arenicola</i> , <i>Holothuria grisea</i> , <i>Holothuria tubulosa</i>	Elongated, squarish, diamond-shaped	Diameter: mean of 6.05 μm , up to 6.90 μm (unspecified shape, including cytoplasm, <i>H. grisea</i>); diameter: mean of 6.95 μm , up to 8.59 μm (unspecified shape, including cytoplasm, <i>H. arenicola</i>); diameter: mean of 12.22 μm , up to 16.74 μm (unspecified shape, including cytoplasm, <i>H. tubulosa</i>)	Queiroz <i>et al.</i> (2022b)
<i>Apostichopus japonicus</i>	No description	Length: 6.7 μm (unspecified shape, determined from image)	Dolmatov <i>et al.</i> (2023)
Unspecified	Rhomboidal, elongated, star-shaped	Length: 4–24 μm (rhomboidal)	Queiroz and Custódio (2024)
Unspecified	Polyhedron	Diameter: 5–10 μm	Xue <i>et al.</i> (2024)
<i>Holothuria scabra</i>	Prismatic, rectangular, elongated	Length: 7–14 μm , up to 25 μm (prismatic); length: ~ 18 μm (fusiform, determined from image), width: ~ 9 μm (fusiform, determined from image)	Wambreuse <i>et al.</i> (2025)
Asteroidea and Holothuroidea (combined)			
Unspecified	Rhomboidal, round	Length: 2–24 μm , usually from 6 to 10 μm (unspecified shape)	Smith (1981)
For example: <i>Cucumaria miniata</i> , <i>Holothuria leucospilota</i> , <i>Patiria miniata</i>	Rhomboidal (or diamond), star-shaped, spherical	Length: 7–8 μm (rhomboidal); width: 4–5 μm (rhomboidal)	Chia and Xing (1996), Xing (1997), Xing <i>et al.</i> (1998)

from a holothuroid and described by Semper (1868) may be a less elaborate form of the star-shaped crystal.

Hexagonal/polyhedron crystals are well documented in a few holothuroids in the genus *Apostichopus* (Dybas and Fankboner, 1986; Xing and Yang, 2008; Xing *et al.*, 2008; Chen *et al.*, 2015; Li *et al.*, 2009, 2018; Liang *et al.*, 2019; Xue *et al.*,

2024). Notably, polyhedrons are hexagonal crystals when viewed from the top and rectangular from the side (Li *et al.*, 2009; Xue *et al.*, 2024). In *A. japonicus*, this is the more commonly encountered crystal shape (Xing *et al.*, 2008; Chen *et al.*, 2015; Li *et al.*, 2018). The diameter of these cells/crystals measure between ~ 5 and 10 μm (Xing *et al.*, 2008; Xing and Yang,

Table 2. Size ranges of crystal cells identified as a type of coelomocyte in echinoderms, re-configured from the morphological summary presented in Table 1 and grouped by crystal shape

Shape and taxon	Length or diameter (μm)	Width (μm)	Reference
<u>Rhomb-shaped</u>			
Asteroidea and Holothuroidea	2–24	No data	Smith (1981)
Asteroidea and Holothuroidea	7–8	4–5	Chia and Xing (1996), Xing (1997), Xing et al. (1998)
Holothuroidea	2–4	No data	Smiley (1994)
Holothuroidea	2–24	No data	Smith et al. (2018)
Holothuroidea	4–24	No data	Queiroz and Custódio (2024)
Holothuroidea	5–6 (mean), 24 (maximum)	4–5 (mean), 16 (maximum)	Hetzel (1963, 1965)
<i>Cucumaria frondosa</i>	15–22	No data	Caulier et al. (2020), Jobson et al. (2021), (2022)
<i>Holothuria glaberrima</i>	5.4 (mean)	No data	Ramírez-Gómez et al. (2010)
<i>Holothuria leucospilota</i>	9–11	7–9	Endean (1958)
<i>Molpadia roretzii</i>	6–8	3–5	Ohuye (1936)
<i>Ocnus lactea</i>	22.0–27.8	No data	Théel (1921)
<i>Panningia hyndmani</i>	11.9–30.4	No data	Théel (1921)
<i>Paraleptopentacta elongata</i>	11.9–30.4	No data	Théel (1921)
<i>Parastichopus tremula</i>	15.2	No data	Théel (1921)
<u>Cuboidal–barrel-shaped</u>			
<i>Apostichopus japonicus</i>	5.9	No data	Xing et al. (2008), Chen et al. (2015)
<i>Paracentrotus gaimardi</i>	~5	No data	Queiroz et al. (2022a)
<u>Star-shaped/stellate</u>			
<i>Panningia hyndmani</i>	11.4–19.0	No data	Théel (1921)
<i>Paraleptopentacta elongata</i>	11.4–19.0	No data	Théel (1921)
<i>Parastichopus tremula</i>	11.4	No data	Théel (1921)
<i>Psolus chitonoides</i>	11.9		Hetzel (1963, 1965)
<i>Psolus</i> spp. and <i>Psolidium</i> spp.	70 (maximum)	No data	Hetzel (1963, 1965)
<u>Hexagonal/polyhedron</u>			
Holothuroidea	5–10	No data	Xue et al. (2024)
<i>Apostichopus japonicus</i>	~5–8	No data	Xing et al. (2008), Xing and Yang (2008), Chen et al. (2015), Li et al. (2018)
<u>Rectangular/prismatic</u>			
Holothuroidea	6–7	4	Eliseikina and Magarlamov (2002)
<i>Holothuria scabra</i>	7–14, 25 (maximum)	No data	Wambreuse et al. (2025)
<i>Mesothuria intestinalis</i>	7.6–15.2	No data	Théel (1921)
<u>Fusiform</u>			
<i>Holothuria scabra</i>	~18	~9	Wambreuse et al. (2025)
<u>Styloid</u>			
<i>Apostichopus japonicus</i>	15.6	No data	Xing et al. (2008), Chen et al. (2015)
<u>Unspecified shape</u>			
Asteroidea and Holothuroidea	6–10	No data	Smith (1981)
<i>Apostichopus japonicus</i>	3.6–10	No data	Li et al. (2009), Li et al. (2013)), Liang et al. (2019), Dolmatov et al. (2023)
<i>Holothuria arenicola</i>	6.95 (mean), 8.59 (maximum)	No data	Queiroz et al. (2022b)
<i>Holothuria grisea</i>	6.05 (mean), 6.90 (maximum)	No data	Queiroz et al. (2022b)
<i>Holothuria tubulosa</i>	12.22 (mean), 16.74 (maximum)	No data	Queiroz et al. (2022b)

2008; Li *et al.*, 2009, 2018; Chen *et al.*, 2015; Xue *et al.*, 2024; Table 2).

Biconvex rhomboid/fusiform/spindle-shaped crystals are known in asteroids (e.g., *Patiria minata*) and holothuroids (Théel, 1921; Andrew, 1962; Johnson and Beeson, 1966). An image taken by Wambreuse *et al.* (2025) may be depicting the fusiform morphology that was referenced by Andrew (1962). This morphology of crystalline structure, isolated from *Holothuria scabra*, measures ~18 µm long and ~9 µm wide (Wambreuse *et al.*, 2025; Table 2).

Crystal cells with rod-shaped/elongated/styloid/needle-like crystals have been noted in asteroids (e.g., *P. minata*) and holothuroids (e.g., the genera *Apostichopus* and *Holothuria*) (Dybas and Fankboner, 1986; Johnson and Beeson, 1966; Xing *et al.*, 2008; Chen *et al.*, 2015; Queiroz *et al.*, 2022b; Queiroz and Custódio, 2024; Wambreuse *et al.*, 2025). In *A. japonicus*, crystal cells with this shape are 16 µm in length (Xing *et al.*, 2008; Chen *et al.*, 2015; Table 2). Rectangular crystalline structures have been observed in the holothuroids *A. japonicus* and *H. scabra* (Eliseikina and Magarlamov, 2002; Wambreuse *et al.*, 2025). Eliseikina and Magarlamov (2002) reported crystal cells with this form as having lengths of 6–7 µm and widths of 4 µm in holothuroids (e.g., *A. japonicus*, *Cucumaria japonicus*; Table 2). Possibly related to either the rod-shaped or the rectangular morphology, the rectangular rod/quadrangular prism crystal isolated from *Mesothuria intestinalis* measured 8–15 µm long (Théel, 1921; Table 2). Furthermore, squarish crystals are found in the genus *Holothuria* (Queiroz *et al.*, 2022b). This morphology could be a variant of the rectangular crystal or another term referring to the cuboidal shape (e.g., side view of the crystal).

Spherical/round crystalline structures inside crystal cells are among the least known morphologies. They are found in asteroids and holothuroids (Smith, 1981; Chia and Xing, 1996; Xing, 1997; Xing *et al.*, 1998; Liang *et al.*, 2019). To our knowledge, there are no images or information on their size (Table 1).

The sizes of unspecified morphologies of crystalline structures (perhaps several shapes pooled together) have been documented. Smith (1981) reported the lengths of crystal cells to be 2–24 µm with typical sizes between 6 and 10 µm in length (Table 2). This was probably referring to the rhomb-shaped crystals because a later study by a different investigator reported the same range for this morphology (Smith *et al.*, 2018). In holothuroids, crystal cells with crystals of unknown morphology measure 6 µm in diameter on average (up to 7 µm) in *Holothuria grisea*, 7 µm in diameter on average (up to 9 µm) in *Holothuria arenicola*, and 12 µm in diameter on average (up to 17 µm) in *Holothuria tubulosa* (Queiroz *et al.*, 2022b; Table 2). In *H. scabra*, prismatic crystals with lengths 7–14 µm (up to 25 µm) are reported (Wambreuse *et al.*, 2025; Table 2). Liang *et al.* (2019) reported crystal cells of *A. japonicus* to be 5–10 µm in length, which was most likely referring to hexagonal crystals (Table 2). Moreover, images of undetermined morphologies of crystalline structures, isolated from *A. japonicus*, ranged from 4 to 7 µm in length (Li *et al.*, 2013; Dolmatov *et al.*, 2023; Table 2).

Cellular components of crystal cells

Cytoplasm and cell membrane of crystal cells

The hyaline cytoplasm and cell membrane surrounding the crystalline structure have been reported by several authors (Semper, 1868; Théel, 1921; Eliseikina and Magarlamov, 2002; Xing *et al.*, 2008; Queiroz *et al.*, 2022b). Hetzel (1963, 1965) further described the cytoplasm as a thin layer around the crystal and only visible

when viewed from a 'favourable angle'. Moreover, the cytoplasm of cells on a prepared microscope slide may have a tendency to spread a little (Hetzel, 1963; Xing *et al.*, 2008; Queiroz *et al.*, 2022a; Queiroz and Custódio, 2024). Crystal cells with pseudopodia were noted by Hetzel (1963, 1965).

Nucleus of crystal cells

Although a single nucleus enclosed within the crystal cell was first observed by Semper (1868), later investigators remarked that it is not always visible, describing it as altogether absent, often absent, or sometimes absent from the cell (Smith, 1981; Chia and Xing, 1996; Xing, 1997; Xing *et al.*, 1998; Queiroz *et al.*, 2022a, b). Furthermore, Théel (1921) noted that the nucleus was detectable after fixation and staining but not in freshly prepared samples. Hetzel (1963, 1965) further added that the nucleus was difficult to see even after fixation. When detected, the nucleus is described as small, heterochromatic/hyperchromatic, crescent-shaped, round, and or flattened (Ohuye, 1936; Hetzel, 1963, 1965; Smith, 1981; Eliseikina and Magarlamov, 2002; Xing *et al.*, 2008; Li *et al.*, 2009; Chen *et al.*, 2015; Smith *et al.*, 2018; Caulier *et al.*, 2020; Dolmatov *et al.*, 2023). Several authors have commented that the nucleus was often located on one side of the crystalline structure, i.e., eccentric position (Ohuye, 1936; Eliseikina and Magarlamov, 2002; Li *et al.*, 2009; Liang *et al.*, 2019; Dolmatov *et al.*, 2023). Moreover, nuclei of crystal cells have been identified on SEM imagery (Xing *et al.*, 2008; Wambreuse *et al.*, 2025).

Other organelles of crystal cells

Organelles other than the nucleus have only occasionally been documented in crystal-cell coelomocytes (Smith, 1981). For instance, rare small vacuoles between 0.03 and 0.25 µm in diameter, occurring in the cytoplasm and next to the cell membrane, were reported by Eliseikina and Magarlamov (2002). A rectangular vacuole (presumably containing a crystalline structure of the same shape) was reported to take up most of the space within the cell (Eliseikina and Magarlamov, 2002; Smith *et al.*, 2018). One to two yellow refractile granules, as large as 4 µm in diameter, were observed in crystal cells studied by Hetzel (1963, 1965).

Colour, staining, and micrography of crystal cells

Colour of crystal cells

The colour appearance of crystalline features in the crystal-cell coelomocytes of holothuroids was described as white by Théel (1921). In the holothuroid *Holothuria leucospilota*, the 'central part' of the cells was reported as green (Chia and Xing, 1996; Xing, 1997; Xing *et al.*, 1998). In the asteroid *Patiria minata*, crystals were pale, golden, and somewhat opalescent (Johnson and Beeson, 1966). Cubic crystals of crinoids were described as refringent and colourless or slightly yellow, those of echinoids (e.g., *Strongylocentrotus lividus*) were yellow, and those of ophiuroids were refringent yellow (Cuénot, 1891).

Staining of crystal cells

Unlike the nucleus and other cellular components, crystalline structures are reported to be resistant to vital/cytochemical staining (Théel, 1921; Ohuye, 1936; Hetzel, 1963, 1965; Eliseikina and Magarlamov, 2002; Queiroz *et al.*, 2022a, b; Xue *et al.*, 2024).

Micrography of crystal cells

Although some investigators report difficulties (Queiroz *et al.*, 2022a), crystal cells have been successfully imaged using SEM

(Xing *et al.*, 2008; Xing and Yang, 2008; Li *et al.*, 2009; Chen *et al.*, 2015; Queiroz and Custódio, 2024; Xue *et al.*, 2024). Using SEM, crystal cells appear to have notches on their surfaces (Li *et al.*, 2009; Xue *et al.*, 2024). To date, attempts to image crystal cells using TEM procedures have been largely unsuccessful (Xing *et al.*, 2008; Li *et al.*, 2009; Chen *et al.*, 2015; Wambreuse *et al.*, 2025), with successful visualisation reported in only two studies (Eliseikina and Magarlamov, 2002; Dolmatov *et al.*, 2023).

Behaviour of crystal cells

Mobility of crystal cells

The lack of mobility of crystal cells has been widely documented (Ohuye, 1936; Chia and Xing, 1996; Xing, 1997; Xing *et al.*, 1998), though some observations of slight movements have also been reported (Théel, 1921).

Osmoregulation by crystal cells

The formation of crystals inside a vacuole in living cells of holothuroids under conditions of increasing osmotic pressure was described by Eliseikina and Magarlamov (2002). They concluded that this process was reversible, supporting the role of crystal cells in osmoregulation, i.e., normalising the osmotic pressure in the body. Under slight osmotic stress, the crystalline structure and the cell containing it, however, had been observed to dissolve and collapse, respectively (Hetzel, 1963). Under laboratory conditions, the concentration of crystal cells did not change when the holothuroid *Cucumaria frondosa* was exposed to low salinities; however, there was an uptick in cell numbers when the holothuroid was exposed to cold air (Jobson *et al.*, 2021). These experimental results downplay the role that these cells play in regulating osmotic pressure (Jobson *et al.*, 2021).

Presence of crystal cells in response to stress

The abundance of crystal cells increased by $100 \pm 122\%$ in the Polian vesicle in response to exposure to cold air of 5°C experienced by the holothuroid *Cucumaria frondosa* (Jobson *et al.*, 2021). Furthermore, the number of crystal cells increased slightly in sea cucumbers exposed to cold air of 17°C and low salinity of 22 psu, but not to low salinity of 15 psu (Jobson *et al.*, 2021).

Chemistry of crystal cells

Solubility of crystal cells

Crystal cells were observed to be insoluble in potassium hydroxide, which is a strong base with a pH from 10 to 13; the exposure time, however, is not specified (Semper, 1868).

Chemical composition of crystalline structures inside crystal cells

Based on the morphology of crystalline structures, Semper (1868) initially hypothesised that these structures were crystals of uric acid. However, he concluded that their insolubility in potassium hydroxide suggested otherwise. In the more recent literature, the chemical composition of crystalline structures inside crystal cells is reported as being largely unknown (Endean, 1966; Smiley, 1994; Queiroz and Custódio, 2024).

Bodily distribution of crystal cells

Derivation of crystal cells

Théel (1921) and Ohuye (1938) posited that crystal cells were derived from lymphocytes, which in turn were derived from coelomic epithelium (also see Kindred, 1924; Hetzel, 1965). In the asteroid *Patiria miniata*, production of 'crystal bodies' was observed from tissue representing the coelomic epithelium; however, it is unknown which cell type from which crystal cells were derived (Johnson and Beeson, 1966). Unlike other types of coelomocytes, crystal cells did not incorporate any injected carbon particles inside the cell, suggesting that they are not capable of phagocytosis of foreign particles and are already highly differentiated (Hetzel, 1965; Zavalnaya *et al.*, 2020).

Occurrence of crystal cells

In holothuroids, crystal cells are commonly found in coelomic fluids (i.e., in the perivisceral and hydrovascular coeloms) and entirely absent from the connective tissue compartment of the body wall (Smiley, 1994; Eliseikina and Magarlamov, 2002; Caulier *et al.*, 2020; but see Ramírez-Gómez *et al.*, 2010). They are also present in aggregates of coelomocytes in the holothuroid *Cucumaria frondosa* (Caulier *et al.*, 2020); however, Dybas and Fankboner (1986) did not find any crystal cells trapped in aggregates (or brown bodies) of the holothuroid *Apostichopus japonicus*. Crystal cells are also known from the coelomic fluids of the asteroid *Patiria miniata* (Johnson and Beeson, 1966). In crinoids, they are found near the oesophagus and in the part of the visceral sac beneath the oral disc (Cuénot, 1891).

Proportion of crystal cells among coelomocytes

The proportion of crystal cells relative to other types of coelomocytes is generally low in holothuroids, ranging from 0.4 to 4.6% (Hetzel, 1963; Queiroz *et al.*, 2022b; Wambreuse *et al.*, 2025). In particular, densities are reportedly greater in the perivisceral than in the hydrovascular fluids even though not all individuals contain crystal cells (Wambreuse *et al.*, 2025). However, in the holothuroid *Apostichopus japonicus*, no difference in crystal cell abundance was detected between perivisceral and hydrovascular fluids (Li *et al.*, 2013).

Factors affecting abundance and size of crystal cells

Density and size of crystal cells were reportedly the same between small (~ 1 cm long) and larger individuals of the holothuroid *Psolus phantapus* (Théel, 1921). Moreover, the size of crystal cells reportedly varied within the same individual (Hetzel, 1963, 1965), while their density tended to decrease over the course of the day (Hetzel, 1963). In *Apostichopus japonicus*, densities of crystal cells did not vary between wild and cultured individuals or among individuals of different sizes (Li *et al.*, 2013; Liang *et al.*, 2019).

Reassessment of crystal cells in holothuroids

Resemblance with microcrystals of uric acid

Upon re-examination of crystal-bearing structures, traditionally defined as crystal cells, isolated from coelomic cavities of the holothuroids *Cucumaria frondosa* and *Psolus fabricii*, a variety of microcrystals were recorded (Figure 1A–I) and convincingly identified as uric acid based on their morphology, and optical and chemical properties (Ma *et al.*, 2025). A summary of the parallels between these uric acid microcrystals and so-called crystal cells is provided below.

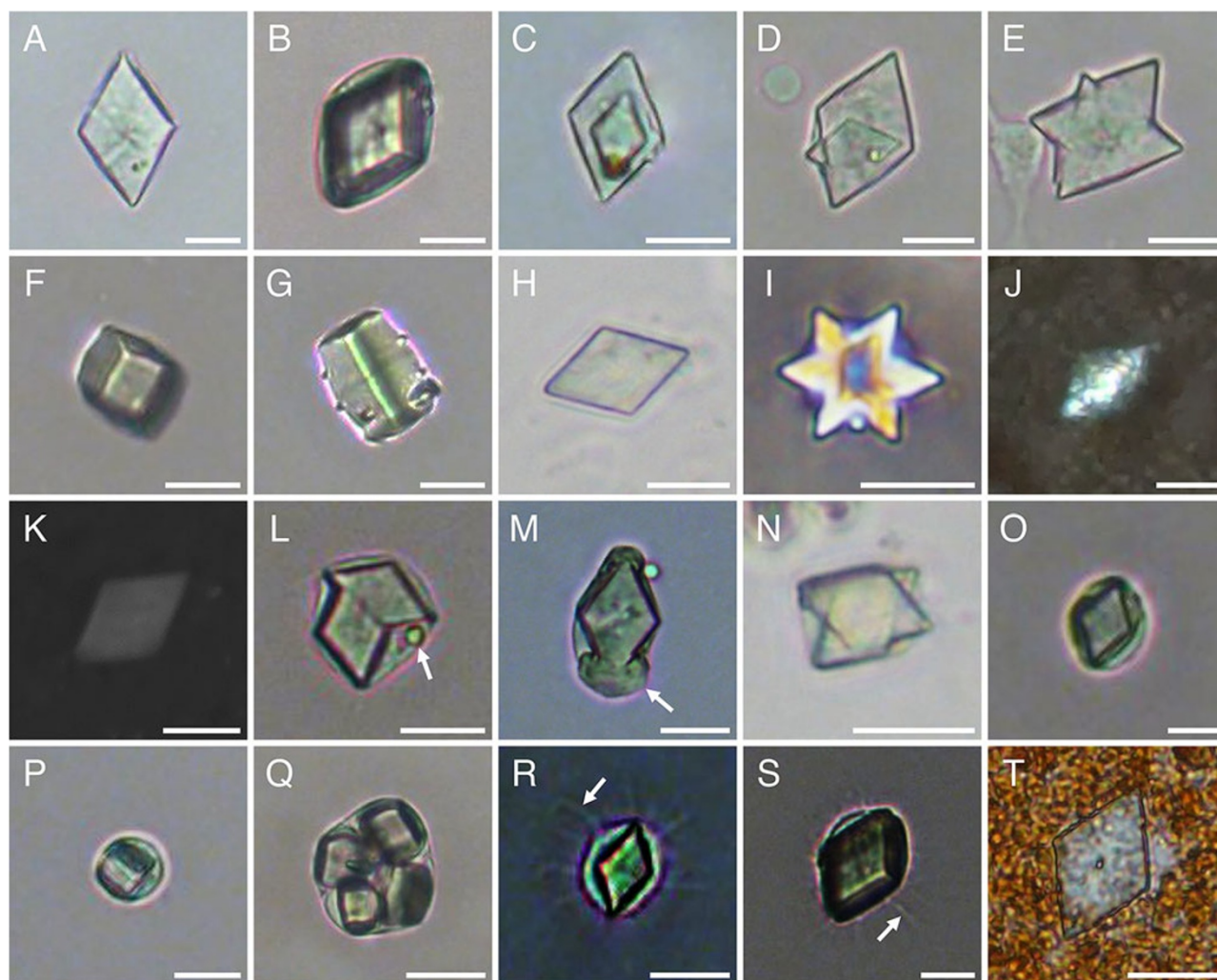


Figure 1. Microcrystals of uric acid isolated from holothuroids: (A) a rhombic crystal, (B) a rhombohedral crystal, (C) two rhombic crystals forming a multi-layered structure, (D) a smaller rhombic crystal growing from a larger one, (E) an irregular star-shaped crystal, (F) a cuboidal crystal, (G) barrel-shaped crystal, (H) a rhombic crystal, (I) three rhombic crystals overlayed at offsetted angles forming a six-pointed star-shaped structure, (J) a rhombic crystal under polarised light, (K) a rhombic crystal imaged using SEM, (L) two rhombic crystals and a foreign particle (arrow) encapsulated in a coelomocyte, (M) a rhombic crystal and amorphous substance (arrow) encapsulated in a coelomocyte, (N) two rhombic crystals encapsulated in a coelomocyte, (O) a single cuboidal crystal encapsulated in a coelomocyte, (P) a single cuboidal crystal encapsulated in a coelomocyte, (Q) four cuboidal crystals encapsulated in a coelomocyte, (R) a rhombic crystal encapsulated in a coelomocyte with filipodia (arrow), (S) a rhombohedral crystal encapsulated in a coelomocyte with filipodia (arrow), and (T) a rhombic crystal trapped in an aggregate of haemocyte coelomocytes. Microcrystals illustrated in panels A–G, J–M, and O–T were isolated from *Cucumaria frondosa* and those in panels H, I, and N were from *Psolus fabricii*. Scale bars represent 10 µm.

Morphological properties

The shapes and sizes of uric acid microcrystals are consistent with those of the crystalline structures ascribed to ‘crystal cells’ (Tables 1 and 2). For instance, the lengths of rhomb-shaped microcrystals in *Cucumaria frondosa* (i.e., 5–29 µm; Ma *et al.*, 2025) and *Psolus fabricii*, and the diameter of a thin star-shaped in *P. fabricii* (a variant of the rhombic shape; e.g., Hetzel, 1963) remains largely within the reported range of rhomb-shaped crystal-cell coelomocytes (Table 2). Furthermore, the width of these crystals in *C. frondosa* (i.e., 4–20 µm; Ma *et al.*, 2025) also aligns closely with the reported range of ‘crystal cells’ (Table 2).

Additionally, the width-to-length ratio of rhomb-shaped microcrystals of uric acid in *C. frondosa* had a mean of 0.60 and a maximum of 0.69 (derived from data collected by Ma *et al.*, 2025), in line with the width-to-length ratios of rhomb-shaped crystal-cell coelomocytes (described above) estimated to average 0.66, based on an analysis of published values, i.e., 0.50–0.63

(*Molpadia roretzii*; Ohuye, 1936); 0.57–0.63 (Chia and Xing, 1996; Xing, 1997; Xing *et al.*, 1998); 0.60 (*P. fabricii*); 0.67, using maximum values (Hetzel, 1963, 1965); and 0.78–0.82 (*Holothuria leucospilota*; Endean, 1958, 1966).

Optical properties

Rhomb-shaped microcrystals of uric acid are colourless in the holothuroid *Cucumaria frondosa* (Ma *et al.*, 2025), but they can exhibit some colouration depending on the solution and surrounding material. Accordingly, the variable colours of ‘crystal cells’ reported in the literature (see above) may also be influenced by the tint of the solution in which they were prepared or the lighting conditions during observation. Moreover, rhombic crystalline structures isolated from the holothuroid *C. frondosa* were optically active under polarised light (Figure 1J), which is a known property of uric acid microcrystals. Finally, difficulties reportedly encountered when trying to image rhombic forms of uric acid crystals with

SEM (Figure 1K) find echoes in some studies of 'crystal cells' (e.g., Queiroz *et al.*, 2022a).

Chemical properties

Solubility tests and Raman spectroscopy of cuboidal–barrel-shaped and rhomb-shaped microcrystals isolated from *Cucumaria frondosa* (Ma *et al.*, 2025), which were previously identified as crystal-cell coelomocytes (Caulier *et al.*, 2020; Jobson *et al.*, 2021), provided more precise information on their chemical properties, which are consistent with those of uric acid. Although chemical analyses have so far focused on these two common morphologies, other crystal forms observed in echinoderms may represent additional morphological variants of uric acid (e.g., rosette-shaped; Kleeberg *et al.*, 1981) crystallised under different physicochemical conditions or, possibly, crystals composed of other chemical compounds.

Evidence of phagocytosis of uric acid microcrystals by coelomocytes

Similar to reports of 'crystal cells', rhomb-shaped, cuboidal, and barrel-shaped microcrystals of uric acid in the holothuroid *Cucumaria frondosa* were found in the perivisceral and hydrovascular fluids; however, the same microcrystals were also detected in the tissues of some internal organs (Ma *et al.*, 2025). In addition, cuboidal and rhombic microcrystals of uric acid were found encapsulated (i.e., in the 'crystal cell' appearance), suggesting that they were phagocytised by a coelomocyte. Small foreign particles may be trapped in the cell (Figure 1L), possibly giving the impression of a nucleus, and an amorphous substance may be present at one or both ends of a rhombic crystal (Figure 1M). Up to two rhomb-shaped crystals of uric acid can be packaged within a single coelomocyte (Figure 1L and N), which matches the reports of crystal-cell coelomocytes with two or more crystals in them (Endean, 1958, 1966; Smith, 1981; Smiley, 1994). Although usually a single crystal is packaged inside a cell (Figure 1O and P), multiple cuboidal crystals, with up to four within a single cell, have been observed in *C. frondosa* (Figure 1Q; Ma *et al.*, 2025). Filopodia were observed on the cell membrane surrounding rhombic (Figure 1R) and rhombohedral microcrystals of uric acid (Figure 1S) (Ma *et al.*, 2025), which is similar to the description of 'crystal cells' with pseudopodia by Hetzel (1963, 1965). Although many of these packaged microcrystals were stationary, a few displayed subtle vibratory/wobbly movements (K. C. K. Ma, personal observation).

Encapsulated microcrystals have been detected in the perivisceral coelomic fluid and the water expelled out of the respiratory tree, which is presumably how they get released from the body (Ma *et al.*, 2025). Another excretion pathway was detected in the form of non-encapsulated crystals of uric acid trapped in the brown-coloured aggregates (Figure 1T), likely in the process of being eliminated through transrectal coelomoducts in the cloaca (Dybas and Fankboner, 1986; Caulier *et al.*, 2020; Jobson *et al.*, 2022). The presence of these excretory naked microcrystals circles back to an observation made a century ago suggesting that the growth of crystalloids in phagocytic coelomocytes caused the degeneration and, eventually, the collapse of the cell (Kindred, 1924).

Conclusion

The detection of crystalline structures with a geometric shape (i.e., prismatic), refringent properties, and resistance to vital staining among coelomocytes of echinoderms gave rise long ago to the

definition of so-called crystal cells in the literature. At one point, they were speculated to represent crystalline structures engulfed by phagocytic coelomocytes in asteroids, but direct observations of the process were lacking to support the hypothesis (Johnson and Beeson, 1966) and it was largely overlooked. Since then, the validity of classifying 'crystal cells' as a distinct type of coelomocyte in echinoderms has been anecdotally called into question a number of times, but never formally examined. Furthermore, the putative role of 'crystal cells' in osmoregulation and ossicle formation has been downplayed due to the lack of direct evidence (Endean, 1966; Ramírez-Gómez and García-Arrarás, 2010; Jobson *et al.*, 2021; see also Diehl 1986, for a review on osmoregulation in echinoderms). Some authors had started referring to them as 'assumed crystal cells' or excluding them entirely from analyses (Carter *et al.*, 2024; Wambreuse *et al.*, 2025). Following further investigation of their morphology and polymorphism, optical activity under polarised light, and solubility, these crystalline structures were determined to be uric acid in holothuroids (Ma *et al.*, 2025), as described in many other phyla. The distribution of free and encapsulated microcrystals sheds further light on their genesis and elimination from the sea cucumber body (Ma *et al.*, 2025). Taken together, the lines of evidence strongly support that what has been termed 'crystal cells' is better understood as a functional state of phagocytic coelomocytes dedicated to the sequestration and transport of crystalline uric acid wastes. This shift in interpretation not only resolves inconsistencies in the roles attributed to this type of coelomocyte but also provides a clearer understanding of metabolic waste management in echinoderms.

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Ethical standards. Not applicable. This work is based on previously published data, and no new experiments involving sea cucumbers were carried out.

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